

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	Y. Krishna Teju Reddy	Reg. No.:	192411200
Section / Batch:	S10t-C	Date:	8-08-2020

Code of Conduct

(Signature)

I, Y. Krishna Teju Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

(Signature)

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:			50 Marks



Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

- a) What are the main objectives of Computer Vision?
- b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

- a) Define feature extraction.
- b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

- a) What is low-level image processing?
- b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

- a) Define a pixel.
- b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

- a) What is the role of an image sensor?
- b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

- a) Define sampling.
- b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

- a) What is perspective projection?
- b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

- a) Define image processing.
- b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

- a) Explain one application in healthcare.
- b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

- a) Why are pixels considered the basic building blocks of digital images?
- b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1 Computer vision

1. Computer vision enables computers to capture, process and understand images and videos.

a. objectives

⇒ object detection

⇒ image understanding.

⇒ pattern recognition.

b. application

⇒ face recognition.

⇒ Medical imaging.

2. Features of computer vision

It extracts useful information from images for automatic analysis.

a. feature extraction: Identifying important image features like edges, corners and textures.

b. object Recognition: Helps identify and classify objects accurately.

3. Levels of Image processing.

Image processing has low-level, mid-level, and high level operations.

a. low level: Image enhancement and noise removal.

b. difference:

Mid level: feature extraction and segmentation.

High level: object recognition and scene understanding.

4. Digital Image fundamentals.

A digital image is made up of pixels arranged in rows and columns.

a. pixel: The smallest unit of a digital image.

b. Resolution: Higher resolution gives clearer and sharper images.

5. Image Sensing and Acquisition.

It converts a real world scene into a digital image using sensors:

a. Image sensor: Converts light into electrical signals.

b. Devices:

→ Digital camera.

→ Scanner.

6. Sampling and Quantization

Ty. They convert analog images into digital form.

a. Sampling: Divides an image into pixels.

b. Quantization: Assigns intensity values to pixels.

7. Image formation model.

It explains how a camera captures an image.

a. Perspective Projection: Converts a 3D scene into a 2D image.

b. Illumination: Provides light for image capture.

Image Processing vs Computer vision.

Image processing improves images, while computer vision understands them.

- a. Image processing : Enhances and modifies image.
- b. Computer vision analyzes and interprets images.

9. Application of Computer vision.

Computer vision is used in many real-world fields.

- a. Healthcare :- Detects diseases from medical images.
- b. Autonomous vehicles : Detects roads, traffic signs and obstacles.

10. Digital Image fundamentals.

Digital image concepts help computer vision analyze images accurately.

- a. pixels : They are the basic building blocks of images.
- b. Sampling & Resolution : Improve image quality and analysis accuracy.